Capstone Project Concept Note and Implementation Plan

Project Title: Screen2Sleep

Team Members

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Concept Note

1. Project Overview

Screen2Sleep AI is a proposed Android-based, privacy-conscious digital well-being assistant designed for university students and young adults aged 18 or older, particularly those with a relatively fixed morning wake-up time. The project addresses bedtime procrastination associated with prolonged late-night smartphone use, especially social media and short-form video sessions. Such sessions can reduce the time available for sleep; use close to bedtime may also be associated with longer sleep-onset latency. The system is a preventive, user-directed support tool, not a diagnostic instrument or a treatment for digital addiction or sleep disorders.

Its key contribution is to translate an ongoing smartphone session into an understandable, personalized estimate of the sleep opportunity that may be at risk. Model A predicts a plausible range for how much longer the current app session may continue, conditional on elapsed session time and context. Model B estimates a possible additional delay in sleep onset associated with bedtime-adjacent use, relative to the user's own baseline. The interface combines (i) projected usage extending beyond a self-selected target bedtime and (ii) the model-based increment in sleep-onset latency. These are estimates with uncertainty, not measured sleep loss or a demonstrated causal effect; the two models are trained on different participant datasets and linked through a defined bedtime-use-duration variable rather than joined at the individual level.

When a projected session may put the user's bedtime at risk, the prototype will offer a brief, explainable just-in-time adaptive intervention (JITAI) that the user can dismiss or suppress for an important session. The initial design tests a monitoring window beginning 60 minutes before target bedtime, with no intervention notifications after that bedtime; the window and decision threshold are design parameters to be evaluated in a pilot, not empirically established optima. The aim is to support informed self-regulation without imposing app blocks or adding further screen engagement.

Sustainable Development Goal (SDG) relevance: The primary alignment is with SDG 3 (Good Health and Well-Being), particularly Target 3.4, which emphasizes promoting mental health and well-being. The project aims to encourage healthier bedtime-related digital habits and awareness of possible sleep disruption. It may also indirectly support SDG 4 (Quality Education) by protecting conditions conducive to students' learning and concentration, but educational or clinical benefits will not be claimed unless evaluated. The expected capstone outcome is a technically feasible prototype and evidence from model validation and limited usability testing, not a proven improvement in sleep health.

2. Objectives

The capstone has the following specific, testable objectives:

- Define and measure the target behavior: reconstruct foreground app-use sessions and usage time around each participant's self-selected bedtime; distinguish directly observed delay past bedtime from the uncertain, modeled sleep-onset component. Focus on adults (18+) rather than labeling users clinically "addicted" on the basis of screen time alone.
- Develop and benchmark Model A: estimate the remaining duration of an ongoing app-specific session from elapsed time, app/context signals, and usage history, using LSApp for initial research and Android usage events for local personalization. Compare with a simple per-app median-duration baseline, and evaluate duration error, prediction-interval coverage, and robustness to incomplete events.
- Develop and benchmark Model B: use available GLOBEM research variables to examine the within-person association between pre-bedtime usage and sleep-onset latency (or, if that target is unreliable, a clearly defined secondary sleep measure). Compare with a personal mean/baseline and examine missingness, device measurement error, temporal confounding, and uncertainty. A team member has submitted the PhysioNet access application and completed the required application steps. GLOBEM access approval is currently pending. Model training using restricted GLOBEM data will begin only after authorization is granted and the applicable data-use conditions are satisfied.
- Create an interpretable early-warning experience: combine the models through a shared bedtime-use-duration variable, show projected bedtime overrun and possible additional sleep-onset delay in minutes or ranges, and explain the assumptions and uncertainty in non-alarmist language. Do not present specific sleep-stage reductions, guaranteed sleep loss, or precise percentages without direct supporting validation.
- Implement user-controlled Android operations: request only the necessary usage access; collect optional sleep information through a short morning diary and—when available and separately authorized—through Health Connect; support a low-battery passive mode and a limited evening active mode; and allow users to silence or disable interventions. Avoid sending intervention notifications after the target bedtime.
- Evaluate feasibility, safeguards, and potential impact: test both models against transparent baselines with participant-wise/temporal splits as appropriate; evaluate reliability across users and device configurations; and use a small, voluntary adult pilot to measure event capture, warning acceptability, false-alarm burden, and the feasibility of sleep data collection. Any improvement in nighttime phone use or sleep will be reported only if the pilot design and observed data support it.

3. Background

Late-evening smartphone use can delay bedtime and reduce the time available for sleep among university students and young adults. Experimental evidence shows that reading on a light-emitting device before bedtime delayed sleep onset and reduced next-morning alertness compared with reading a printed book (Chang et al., 2015). In a cross-sectional survey of adults, mobile phone use after lights out was associated with longer sleep latency, poorer sleep efficiency and greater fatigue, with the fatigue association found among younger adults (Exelmans & Van den Bulck, 2016). Objectively logged screen-time around bedtime was likewise associated with longer sleep-onset latency and poorer sleep quality in a cross-sectional study of adults, although reverse causation could not be excluded (Christensen et al., 2016). Reviews in children and adolescents report similar associations between screen time, delayed bedtimes and shorter sleep (Cain & Gradisar, 2010; Hale & Guan, 2015). The direction of the relationship is not settled: a recent theoretical review proposes bidirectional links, in which sleep problems may also precede technology use, for example when devices serve as a time filler or an emotion-regulation strategy during the sleep-onset process, and notes that the association may differ across individuals (Bauducco et al., 2024). Screen2Sleep therefore treats its outputs as estimates of sleep opportunity at risk, not as causal sleep loss, and supports SDG 3 (Target 3.4) through awareness and self-regulation rather than treatment.

Existing tools, such as built-in digital well-being features (Google, n.d.), mostly rely on fixed daily limits or scheduled reminders. Reviews of digital self-control tools note that such designs differ widely in how well they support users' own goals (Lyngs et al., 2019). In a six-week field experiment, a self-nudging app that adds a short pause and an option to dismiss when a target app is opened reduced users' app openings, with the dismiss option having the strongest effect (Grüning et al., 2023); adaptive and explainable designs have also been proposed (Orzikulova et al., 2024), in line with the just-in-time adaptive intervention (JITAI) framework (Nahum-Shani et al., 2018). A machine learning approach is useful because the right moment to intervene depends on how long an ongoing session is likely to continue, which varies by app, context and person. Session length is positive, skewed and often right-censored, so survival (time-to-event) models are better suited than ordinary regression (Cox, 1972; Vasiloudis et al., 2017). Personalized estimates of the link between pre-bedtime use and sleep onset also require longitudinal, within-person modelling.

4. Methodology

Screen2Sleep will use two complementary models: Model A will estimate the remaining duration of an ongoing app session, while Model B will investigate the association between evening smartphone use and subsequent sleep outcomes. Data preparation, training and evaluation will be performed locally in Python using pandas, NumPy, lifelines, XGBoost and statsmodels.

For Model A, app-level usage intervals will be reconstructed from LSApp events. Duplicate events, invalid durations and broken sequences will be identified using documented cleaning rules. An incomplete session will be treated as right-censored only when a reliable minimum duration can be established. We will compare a Weibull accelerated failure time (AFT) model with a Cox proportional hazards model and an XGBoost-AFT benchmark. Survival analysis allows session duration to be modelled while accounting for censored observations (lifelines, n.d.; XGBoost Developers, n.d.). Candidate predictors will include app identity, weekday and prior usage history; time-of-day features will be included only if timestamp interpretation is verified. At runtime, predictions will be conditioned on the time the session has already lasted, producing a median remaining duration and a prediction interval.

For Model B, we propose a linear mixed-effects model with a participant-specific random intercept to account for repeated observations from the same person (statsmodels Developers, n.d.). Evening usage will be separated into the participant's usual usage level and deviations from that level, allowing within-person and between-person associations to be examined separately. Available contextual variables, such as weekday and study cohort, will also be considered. Sleep-onset latency will be used only if the relevant fields contain sufficiently complete and meaningful measurements. Otherwise, a separately defined outcome, such as sleep duration, will be evaluated and clearly labelled; it will not be interpreted as additional time needed to fall asleep.

Evaluation will distinguish performance on unseen users from prediction of future observations for existing users. Training and test partitions will respect participant identities and temporal order. Preprocessing statistics and personal baselines will use training or previously observed data only. Model A will be compared with an elapsed-time-conditioned median remaining-duration baseline; Model B will be compared with a historical personal-mean baseline, using a training-population baseline for new users. We will report mean absolute error and median absolute error for observed outcomes, together with prediction-interval coverage and width. Censored sessions will additionally require censoring-aware evaluation.

LSApp and GLOBEM will be analysed separately because they contain different participants. Any connection between the models will require compatible usage definitions, time windows and units, followed by validation on pilot data. Full-evening usage summaries will not be assumed to represent usage immediately before an individual's bedtime. Model A will initially support a projected bedtime-overrun warning. A numerical sleep-related estimate will be added only if Model B and the connection between the models are sufficiently validated. All outputs will be presented as uncertain predictions or associations, rather than causal estimates of sleep loss.

5. Architecture Design Diagram

Screen2Sleep AI separates **offline training** from **on-device runtime** (Figure 1). Two research datasets, with credentialed access required for GLOBEM are used on a local machine to fit population-level starting models: LSApp for Model A (remaining duration of an ongoing app session) and GLOBEM for Model B (association between pre-bedtime use and sleep-onset latency). Only compact model parameters, never raw records, move from the offline side to the Android application. On the phone, every step from usage logging to inference, warning and personalisation runs locally; there is no application server. The two models are trained on different people and are never merged row-wise; they communicate through a single shared interface variable, pre-bedtime use relative to the person's own evening norm.

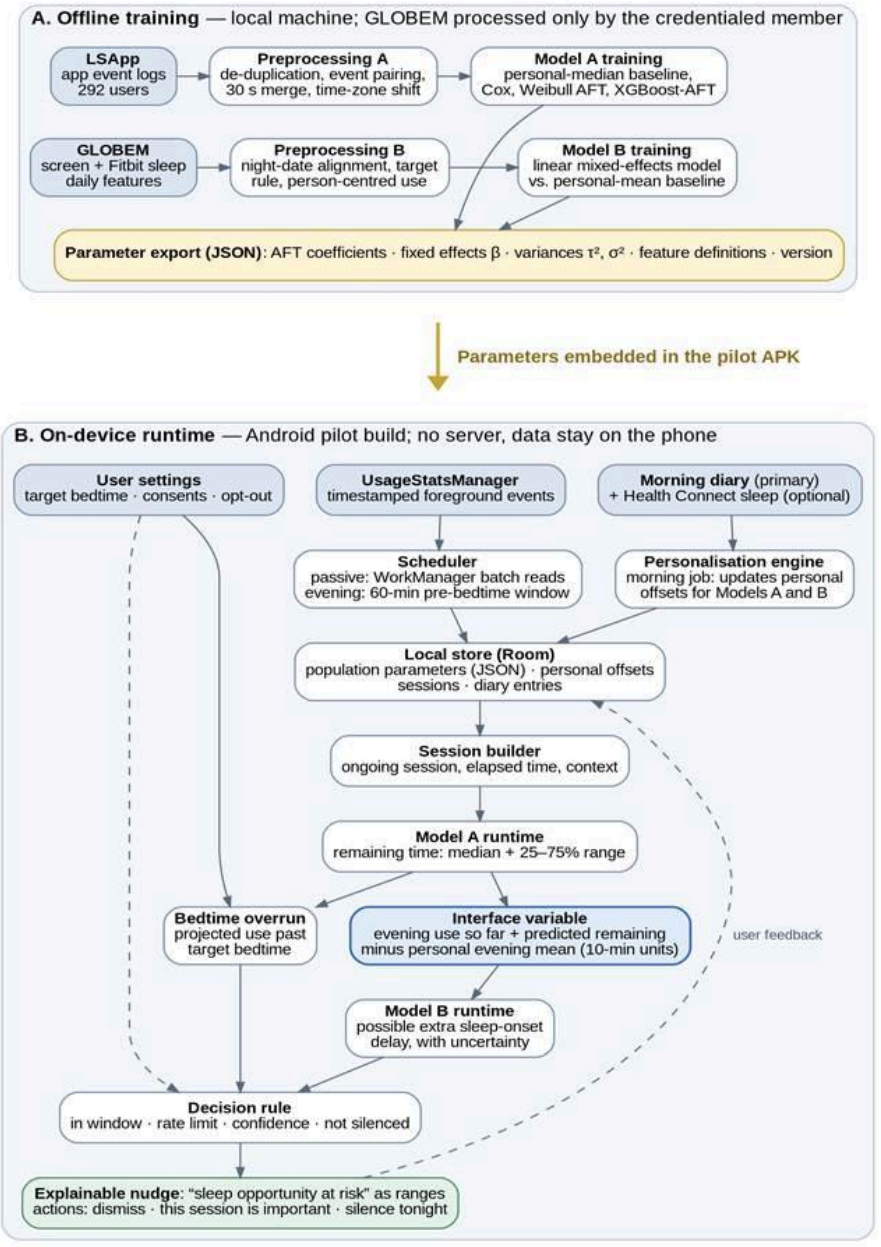


Figure 1. Screen2Sleep AI architecture. Panel A: offline training of Model A (LSApp) and Model B (GLOBEM) on a local machine. Panel B: on-device runtime on Android, where the two models are linked through one interface variable and personalised with the user's own data.

Component descriptions

Component	Role and functionality
A. Offline training (local machine)	
Open datasets	LSApp supplies timestamped app events from 292 Android users and trains Model A. GLOBEM supplies daily phone-use and Fitbit sleep features from the same students and trains Model B. GLOBEM is used only after PhysioNet approval and is processed solely on the credentialed member's computer, without cloud or AI services.
Preprocessing A	Rebuilds app-level sessions from raw events: consecutive duplicates removed, Opened/Closed events paired per session and app, re-entries within 30 s merged, timestamps shifted from UTC to local time. The result is 213,028 sessions; unmatched sessions are kept as right-censored observations.
Preprocessing B	Follows the pre-registered GLOBEM plan: verifies which night each sleep record belongs to, selects the target (sleep-onset latency, or time awake in bed if latency is zero on most nights) and centres evening (18:00–24:00) screen time on each person's own mean.
Model A training	Predicts the remaining time of sessions that have already lasted 5 minutes. Candidates are the personal-median baseline, a Cox model, a Weibull accelerated failure time (AFT) model and XGBoost-AFT, evaluated with chronological and user-held-out splits. The AFT model is the deployment candidate; XGBoost-AFT stays an offline benchmark unless it clearly outperforms it (milestone M4).
Model B training	Linear mixed-effects model: target ~ within-person evening use (10-min units) + between-person evening use + weekend + cohort, with a person-specific random intercept (random slope if it converges). The baseline to beat is the person's mean sleep-onset latency.
Parameter export	Writes a small JSON file with AFT coefficients, Model B fixed effects and variance components (τ^2, σ^2), feature definitions and a version tag. GLOBEM-derived parameters are included only in the non-public pilot build; any public release would start from literature-based values instead, in line with Section 5 (Ethical Considerations).
B. On-device runtime (Android)	
UsageStatsManager	Provides timestamped foreground events once the user grants usage access. A revoked permission or unreadable log is recorded as missing data, never as zero use.
Scheduler	Passive mode: WorkManager reads new events in batches during the day to build personal history (no notifications). Evening mode: more frequent checks in a 60-minute window before the target bedtime; if the operating system blocks automatic start, the user starts the mode with one tap.
Local store	Room (SQLite) database in app-private storage holding sessions, diary entries, population parameters and personal offsets. It relies on device encryption and can be wiped with one tap.

Session builder	Applies the same session rules as Preprocessing A to live events and computes the model inputs: elapsed time, app or category, time of day, weekday and use so far today.
Model A runtime	Kotlin implementation of the Weibull AFT model. Given the elapsed time, it returns the conditional median remaining time and a 25–75% range in closed form.
Bedtime overrun	Compares the projected end of use with the user's target bedtime. This is the directly observable part of the warning.
Interface variable	Evening use so far plus Model A's predicted remaining time, minus the person's usual evening use, in 10-minute units. This matches the person-centred predictor in the GLOBEM analysis plan; expressing use relative to the person's own norm reduces scale differences between the datasets. Its transportability is checked in the pilot.
Model B runtime	Combines population fixed effects with the person's offset to estimate a possible additional sleep-onset delay, with an uncertainty range. It is presented as an approximate association, not a causal effect.
Personalisation engine	Morning WorkManager job. Takes the morning diary (primary measure, adapted from the Consensus Sleep Diary) and, if authorised, Health Connect sleep sessions, then updates the personal offsets with a Bayesian normal update (see Implementation Plan, Section 1). With few nights the estimate stays close to the population value; it moves toward the person as valid nights accumulate. No fixed personalisation period is assumed.
Decision rule	Shows a nudge only if all conditions hold: inside the evening window and before bedtime, frequency cap not reached, enough data and an acceptably narrow prediction range, and the user has not silenced the session or evening. Otherwise the system stays silent.
Explainable nudge	One short, neutral notification expressed as ranges, e.g. "This session often lasts another 15–30 minutes, which may take you past your bedtime." Actions: dismiss, "this session is important", silence tonight. Responses are stored locally for evaluation.
User settings	Target bedtime, separate consents (usage access, notifications, Health Connect), opt-out and data deletion.

Graceful degradation. Each layer can fail without breaking the system. If GLOBEM access or label quality does not allow a reliable Model B, the nudge falls back to the bedtime-overrun estimate alone; if Model A does not beat the personal-median baseline, the baseline is used for the range; if no wearable is available, the morning diary provides the sleep label.

6. Data Sources

Screen2Sleep will use LSApp for session-duration modelling and GLOBEM for exploratory modelling of smartphone use and sleep. LSApp contains 599,635 app-usage records from 292 participants, including user identifiers, session identifiers, timestamps, app names and event types (Aliannejadi, n.d.). Preprocessing will reconstruct app-level intervals, remove duplicate or invalid records, assess incomplete sequences and verify timestamp semantics. GLOBEM contains longitudinal smartphone

and wearable-derived features from 497 unique participants across four annual studies (Xu et al., 2022a, 2023). Its screen and sleep feature files will be inspected for compatible usage measures and usable sleep outcomes. Preprocessing will align usage with the corresponding sleep episode, assess missingness and measurement quality, and calculate personal reference values from past observations only. GLOBEM's fixed evening window will not be treated as an exact individual pre-bedtime window. The datasets will not be merged at participant level. Optional pilot data will include locally recorded Android usage events and morning sleep-diary responses for feasibility testing and personalisation. Restricted GLOBEM data will be used only after the relevant researcher has satisfied the applicable access and data-use requirements.

7. Literature Review

Four strands of literature support Screen2Sleep. First, studies of evening light-emitting device use and bedtime phone use report delayed sleep onset, longer sleep latency and reduced next-morning alertness in experimental and cross-sectional designs (Chang et al., 2015; Exelmans & Van den Bulck, 2016), with similar patterns in reviews of younger populations (Hale & Guan, 2015), although recent theoretical work argues that the relationship is bidirectional and moderated by individual factors (Bauducco et al., 2024). Second, smartphone sensing studies show that objectively logged phone use is associated with sleep and well-being in cross-sectional (Christensen et al., 2016) and longitudinal samples (Wang et al., 2014; Xu et al., 2022a). Third, work on app dwell time and session length shows that how long a user stays can be predicted from context and usage history (Tian et al., 2021), and survival analysis with gradient boosted trees has been used to predict session length in music streaming, which suits censored durations (Vasiloudis et al., 2017). Fourth, research on digital self-control tools and just-in-time interventions indicates that well-timed, low-friction prompts that leave the choice to the user can reduce app use and are a promising alternative to static limits (Grüning et al., 2023; Lyngs et al., 2019; Nahum-Shani et al., 2018; Orzikulova et al., 2024). Together, these strands leave several gaps: (1) most studies use aggregate daily screen time rather than session-level remaining duration; (2) existing tools use fixed thresholds instead of probabilistic, personalized prediction; (3) session-duration prediction is rarely connected to bedtime or sleep opportunity; (4) population-level associations say little about within-person sleep responses; and (5) consumer wearable sleep measures are imperfect, with high sensitivity but low specificity for wakefulness (Chinoy et al., 2021; Schyvens et al., 2025), so sleep estimates need explicit uncertainty. Screen2Sleep addresses these gaps with Model A (survival-based remaining-session prediction on LSApp; Aliannejadi, n.d.) and Model B (within-person association between pre-bedtime use and sleep-onset latency on GLOBEM; Xu et al., 2022b), linked through a shared bedtime-use-duration variable, without claiming causal effects.

Implementation Plan

1. Technology Stack

The stack has two parts that share one contract: a Python environment on a local machine for data preparation, model training and evaluation, and a native Android application that runs inference and personalisation on the device. Choices favour small, explainable models that can run cheaply on a phone without sending data to a server.

Category	Technologies and tools	Purpose in Screen2Sleep AI
Programming languages	Python 3.11; Kotlin; SQL (via Room)	Python for analysis and training; Kotlin for the Android app and on-device model code.
Data processing	pandas (McKinney, 2010), NumPy, matplotlib	LSApp session reconstruction, GLOBEM feature preparation, descriptive statistics and figures.
Modelling and evaluation	lifelines (Davidson-Pilon, 2019); xgboost (Chen & Guestrin, 2016); statsmodels (Seabold & Perktold, 2010); scikit-learn (Pedregosa et al., 2011)	Kaplan–Meier, Cox and Weibull AFT models; XGBoost-AFT benchmark; linear mixed-effects model (MixedLM); data splits and error metrics (MAE, median AE, C-index, interval coverage).
Development environment	Jupyter and VS Code (local); Android Studio; Git with a private GitHub repository	Reproducible scripts; the repository holds code only, never LSApp/GLOBEM records or pilot data.
Android framework	Android SDK (minimum API 28, Android 9); Jetpack Compose; Kotlin coroutines	User interface (settings, morning diary, nudge screens) and asynchronous work. API 28 is the lowest level supported by Health Connect.
Android platform APIs	UsageStatsManager / UsageEvents; WorkManager; foreground service (evening mode); notifications (POST_NOTIFICATIONS); Health Connect client (SleepSessionRecord, READ_SLEEP)	Usage logging, batch and scheduled jobs, the evening monitoring window, nudges, and optional wearable sleep data.
Storage and serialisation	Room (SQLite); Jetpack DataStore; kotlinx.serialization	Local database for sessions, diary and parameters; user preferences; reading the exported JSON parameters.
On-device inference	Plain Kotlin (no ML runtime)	Weibull AFT and Model B formulas are a few arithmetic operations, so LiteRT or ONNX Runtime is not required.
Testing	JUnit; Android emulator and physical devices; Android Studio profilers	Unit tests, Python–Kotlin parity tests, battery and latency measurement in the pilot.
Hardware	Team members' own Android smartphones (Android 9 or later, ideally more than one manufacturer); optional wearable that writes sleep to Health Connect; personal laptops	Pilot devices and local analysis. Several manufacturers are needed because battery policies affect background work. No cloud servers or GPUs are required.

Model deployment path

Models are trained in Python and exported as JSON parameters that are packaged with the pilot APK and evaluated in Kotlin. For Model A, the Weibull AFT model gives $\lambda = \exp(\beta \cdot x)$ and a shape

parameter ρ, so the q-th quantile of the remaining time after t elapsed minutes has the closed form r_q = λ·[(t/λ)^ρ − ln(1 − q)]^(1/ρ) − t. The median uses q = 0.5, and the displayed range uses q = 0.25 and 0.75. For Model B, the personal offset is updated each morning with the normal–normal Bayesian rule $\hat{u} = \Sigma e_i / (n + \sigma^2/\tau^2)$, where e_i is the difference between the observed sleep-onset latency and the population prediction on night i, and n is the number of valid nights. The posterior variance, $1 / (n/\sigma^2 + 1/\tau^2)$, provides the uncertainty shown to the user. The same rule, applied to session residuals, updates the personal adjustment of Model A.

Before each pilot build, a parity test runs the same set of test inputs through the Python models and the Kotlin implementation and requires identical outputs within rounding error. Because there is no server, new parameters reach the device only through a new app build, which is acceptable for the pilot. XGBoost-AFT and sequence models such as LSTM or GRU remain offline research options; they would be ported to the device only if they clearly outperform the AFT model and justify the extra size, battery cost and loss of explainability.

2. Timeline

The capstone runs from 12 to 20 October 2026 and is organised into eight phases. Phases overlap where work can run in parallel: the GLOBEM access request and analysis plan run alongside LSApp preprocessing, and documentation starts while evaluation is being completed. Each phase has a lead who owns its deadline and supporting members. Because GLOBEM access requires credentialed approval on PhysioNet, Model B is prepared as a ready-to-run pipeline within the capstone window, and its effect size is estimated once access is granted.

2.1 Gantt Chart (October 2026)

Phase	Main tasks	12	13	14	15	16	17	18	19	20	Lead (support)
0. Kick-off & scope	SDG 3 link, RQ1–RQ4, two-model architecture, task split	■									All
1. Literature & technology review	4 literature themes, 5 research gaps; Android & ML technology review	■	■								Büşra Kabak Yusufcan Özdemir
2. Data access & preprocessing	LSApp cleaning & session building; GLOBEM access request & pre-registered analysis plan	■	■	■	■						Yaren Barış Yusufcan Özdemir
3. Model development	Model A: baseline, Cox/AFT, XGBoost-AFT; Model B: mixed-effects + Bayesian updating; interface variable				■	■	■				Yaren Barış Yusufcan Özdemir

4. Training & evaluation	Time-ordered & user-held-out tests, MAE vs. baseline, interval coverage, subgroup check										Yusufcan Özdemir Hatice Bensu Coşar,
5. Deployment (Android prototype)	UsageStats logging, evening mode, Health Connect, morning diary, on-device inference, alert rules										Yaren Barış, Yusufcan Özdemir
6. Ethics & documentation	KVKK consent texts, risk table, merge sections, references										Hatice Bensu Coşar, İsra Deniz Çandır
7. Final submission	Final review and submission										İsra Deniz Çandır

2.2 Task Breakdown and Deadlines

ID	Task	Start	Deadline	Owner
T1	Kick-off: agree on problem, target group (18+ students/young adults), SDG 3 and RQ1–RQ4	12 Oct	12 Oct	All
T2	Literature review on 4 themes and research-gap analysis	12 Oct	13 Oct	Büşra Kabak
T3	Technology review (UsageStatsManager, WorkManager, Health Connect, survival models)	12 Oct	13 Oct	Yusufcan Özdemir
T4	Download LSApp; submit GLOBEM credentialed access request on PhysioNet	12 Oct	12 Oct	Yaren Barış
T5	LSApp cleaning: remove duplicates, session rules (30 s merge), time-zone alignment	13 Oct	15 Oct	Yaren Barış, yusufcan Özdemir
T6	Compute baselines (personal median remaining time) and define test splits	15 Oct	15 Oct	Yusufcan Özdemir
T7	Write pre-registered GLOBEM analysis plan for Model B	14 Oct	15 Oct	Yaren Barış
T8	Model A: fit Cox, parametric AFT (lifelines) and XGBoost-AFT	15 Oct	17 Oct	Yusufcan Özdemir
T9	Model B: mixed-effects model code, Bayesian prior pipeline, interface variable	16 Oct	17 Oct	Yaren Barış, Yusufcan Özdemir
T10	Evaluate Model A on held-out users: MAE, median AE, C-index, interval coverage	17 Oct	18 Oct	Yusufcan Özdemir
T11	Ethics: KVKK information text, consent flow, risk & mitigation table	17 Oct	18 Oct	Hatice Bensu Coşar
T12	Android prototype: passive logging, evening mode, morning diary, alert rules	18 Oct	19 Oct	Yaren Barış
T13	Merge all sections, unify references (APA 7), task matrix and formatting	18 Oct	19 Oct	İsra Deniz Çandır
T14	Final proofreading and submission	20 Oct	20 Oct	İsra Deniz Çandır

2.3 Task Distribution Matrix

The matrix clarifies each member's responsibility. R = Responsible; C = Contributes. Each member leads roughly an equal share of the work.

Task / Section	Hatice Bensu Coşar	Büşra Kabak	Yusufcan Özdemir	Yaren Barış	İsra Deniz Çandır
Project Overview, Objectives (SDG 3, RQ1–RQ4)	**R**				C
Background, Literature Review (4 themes, gaps)	C	**R**	C		
Methodology, Technology Stack		C	**R**	C	
Architecture Diagram, Data Sources (LSApp, GLOBEM)			C	**R**	
Data preprocessing (LSApp sessions, GLOBEM plan)			C	**R**	
Model A & Model B development and evaluation	C	C	**R**	C	
Android prototype (deployment)			C	**R**	
Challenges & Mitigations, Ethics (KVKK)	**R**		C	C	
Timeline, Milestones (project management)	C	C	C	C	**R**
References, final formatting & submission	C	C	C	C	**R**

3. Milestones

Each milestone has a verifiable completion criterion. They cover task completion, data readiness, performance targets, and key features. The performance target in M4 is tied to the baseline measured on LSApp, so Model A is only kept if it does better than a simple personal-median rule. M8 extends beyond the capstone and tests the system on real users.

#	Date	Milestone	Completion criterion
M1	12 Oct	**Scope locked**	Problem, target group, SDG 3 link, RQ1–RQ4 and the two-model architecture (Model A + Model B linked by one interface variable) agreed; tasks assigned.
M2	13 Oct	**Reviews complete**	Literature review (4 themes, 5 research gaps) and technology review finished; technology stack selected.
M3	15 Oct	**Clean LSApp dataset & baselines**	213,028 valid app sessions after duplicate, session and time-zone rules. Baseline fixed: personal-median MAE of 3.06 min (sessions > 5 min) and 7.15 min (> 15 min). GLOBEM access requested and analysis plan pre-registered.
M4	17 Oct	**Model A beats baseline**	Best Model A (AFT or XGBoost-AFT) has lower MAE than the personal-median baseline on held-out users for both > 5 min and > 15 min sessions; prediction-interval coverage reported.

M5	18 Oct	**Model B pipeline ready**	Mixed-effects model and Bayesian updating code ready to run on GLOBEM; no numeric effect size claimed until the pre-registered analysis is run.
M6	19 Oct	**Android prototype works end-to-end**	Prototype logs sessions (UsageStatsManager), runs the evening mode, collects the morning diary and shows a non-causal alert; no alerts after target bedtime.
M7	20 Oct	**Final submission**	Concept Note and Implementation Plan merged, referenced, proofread and submitted.
M8	After 20 Oct	**Team pilot (2 weeks)**	Pilot with written KVKK consent from team members: battery use, missing-log rate, alert acceptance and morning sleep-onset data collected.

4. Challenges and Mitigations

The following risks are anticipated. The stated controls are proposed safeguards and evaluation activities, not completed or verified implementation outcomes.

- **Event-log quality, inconsistent session definitions and time zones.** LSApp includes event-level records, but one recorded session may contain several apps; missing closures, "Broken" events, and app transitions can distort app-level durations. Reconstruct app-specific foreground intervals with documented pairing rules, distinguish genuinely right-censored events from corrupt records, investigate alternative short-gap merging thresholds, and examine outliers. Verify time zone semantics before creating "time of day" features; otherwise, omit that feature rather than assume a local clock.
- **Limited and uncertain sleep labels.** As access to GLOBEM is currently pending, the team has not yet independently verified the completeness and reliability of its bedtime and Fitbit-derived sleep-onset variables. Once access is approved, missingness, label availability, and data quality will be systematically assessed. Consumer wearable devices may also misclassify quiet wakefulness. Quantify missingness by participant, cohort, time, and platform; avoid silently treating absent data as zero; examine plausible missing-data mechanisms and sensitivity analyses. Retain a brief morning self-report based on established sleep-diary concepts as an important complementary measure. If reliable sleep-onset estimation is not feasible, narrow the output to measurable bedtime overrun or a clearly labeled secondary sleep proxy.
- **Dataset access, permitted use and sample mismatch.** GLOBEM requires authorized access through PhysioNet and compliance with its data-use agreement. One team member has already submitted an access application and completed the required application procedures; approval is pending. Restricted data will only be accessed after authorization is granted to the relevant individual. Credentials and restricted records will not be shared among team members. LSApp licensing and redistribution conditions will also be reviewed before use. LSApp and GLOBEM contain different individuals and must not be row-wise merged. Connect separately trained components only through harmonized, documented usage-duration features; test transportability before presenting a combined prediction.

- **Weak prediction, calibration, and cold start.** Sessions are highly variable, and the relationship between smartphone use and sleep differs across users. Benchmark Model A against per-app median sessions and Model B against a person's ordinary sleep-onset duration. Use subject-aware and chronological evaluation, inspect subgroup errors, report predictive uncertainty, and suppress a numeric warning when quality is inadequate. Begin with population/reference estimates and update user-specific parameters only after sufficient valid observations; a fixed "fully personalized after 2–3 weeks" guarantee is not justified.
- **Correlation, confounding, and overstatement of health effects.** Late use may precede poor sleep, but difficulty sleeping may also lead a person to use the phone; wearables cannot establish physiological causation from app logs. Use within-person comparisons where possible, separate use before bedtime from overnight use, and control for available contextual factors. Label predicted values as approximate associations or sleep opportunity at risk, not causal "minutes of sleep stolen." Do not infer deep-sleep/REM percentages or clinical benefit without appropriate data and study design.
- **Android operating-system and device restrictions.** UsageStatsManager requires user-granted usage access. Detailed events, background execution, notification permissions, battery policies, and foreground-service rules vary by Android version or manufacturer. Use scheduled background work for periodic history collection, not as a guarantee of second-by-second monitoring. Prototype an evening mode compatible with current restrictions (including user-initiated activation if necessary), test on representative devices, and display missing-permission or unavailable-data states rather than mistaking them for zero usage.
- **Notification fatigue, sleep-related anxiety, and low adoption.** Excessive warnings may increase engagement or "orthosomnia"-type preoccupation with sleep metrics. Limit interventions to a short pre-bedtime window; rate-limit and allow snoozing/opt-out; stop messages after the bedtime target; and avoid gamified streaks, blame, red-alert scoring, or coercive language. Measure perceived intrusiveness and false-alarm frequency in the pilot; adjust or disable warnings if negative effects emerge.
- **Sensitive data leakage, unfair performance and limited inclusion.** App histories can reveal personal routines, while sleep information may constitute special-category health data. Minimize variables, keep data on-device wherever feasible, use secure storage and explicit retention/deletion controls, and restrict access to any research data. Assess performance among participants with and without wearables, across phone versions and with varied schedules. Provide a wearable-free morning-diary pathway; clearly state that initial targeting is adults with relatively fixed wake-up times, rather than generalizing to all populations.

5. Ethical Considerations

Privacy, purpose limitation and data minimization. Screen2Sleep AI combines behavioral usage traces with potentially sensitive sleep information. Its prototype should collect only what is required for the stated features—such as timestamps, session duration, app identity or category, a user-selected bedtime and optional sleep-diary entries. It should not access message contents, contacts, browsing content, microphone recordings or precise location. Where possible, features and predictions should be processed and retained locally on the Android device rather than uploaded to a cloud service. Any additional purpose, such as research export, requires a separately assessed basis and transparency.

Consent, transparency and Turkish data-protection obligations. Participants and future users should receive a clear notice explaining which data are processed, why, their retention period, potential recipients and how to exercise applicable rights under Türkiye's Personal Data Protection Law (Law No. 6698; KVKK). Health-related information is special-category personal data; the applicable Article 6 processing condition and safeguards must be established before collection. For the proposed voluntary pilot, obtain specific, informed and withdrawable explicit consent where this is the applicable legal basis; consent must not be bundled with unrelated permissions. A usage-access setting in Android, Health Connect authorization and lawful consent/data-processing conditions are distinct requirements. A qualified review should confirm the final legal basis, controller roles and security procedures before data collection.

Confidentiality and governance. Apply privacy-by-design, access controls, encryption supported by the device, secure development practices, limited retention and straightforward deletion/permission-revocation mechanisms. If a research export is genuinely needed, use a separate, justified protocol with pseudonymization, approved secure storage and strict researcher access; pseudonymization alone does not make records anonymous. Do not put identifiable or restricted records in public repositories, analytics platforms or third-party generative-AI services. Document data deletion at the pilot's conclusion, subject to legally necessary retention. Never attempt to re-identify participants in LSApp or GLOBEM.

Restricted research-data conditions. A team member has applied for credentialed access to GLOBEM through PhysioNet and completed the required application procedures. The team is currently awaiting the access decision. Until authorization is granted, no restricted GLOBEM data will be downloaded, processed or used for model training. Following approval, data access and processing will be limited to individually authorized researchers and conducted in accordance with the applicable data-use agreement, security requirements and permitted research purposes. Credentials and restricted data will not be shared with unauthorized team members. LSApp's licensing conditions will also be reviewed. No restricted datasets or model artifacts with uncertain redistribution rights will be incorporated into the publicly distributed application without appropriate authorization.

Autonomy, non-maleficence and appropriate clinical boundaries. Warnings should explain uncertain estimates in neutral language, offer a genuine choice to continue, dismiss or temporarily silence them, and avoid encouraging additional checking of the app. No intervention notifications are sent after the user's bedtime goal. Screen2Sleep AI must not assign an addiction diagnosis, replace clinical sleep assessment, or present its estimates as medical advice. Persistent sleep problems, significant distress or concerns about loss of control should trigger optional signposting to an appropriate health professional or relevant support services (for example, Yeşilay's YEDAM), rather than automated treatment claims.

Bias, accessibility and responsible evaluation. Predictive performance may differ by age, gender, usage patterns, device type, access to wearables and schedule regularity; restricted or incomplete data could amplify these differences. Evaluate errors and interval calibration across relevant groups where sample sizes allow, avoid ungrounded conclusions about small subgroups, and offer a diary-based option for participants without wearables. Use accessible, understandable language and avoid stigmatizing expressions such as identifying someone as “addicted” from usage counts alone. Findings should report uncertainty, limitations and adverse user feedback as well as positive results.

Pilot-participant protection and accountability. Recruit only consenting adults (18+), including any team-member volunteers, without pressure or penalties for declining. Explain the prototype's experimental status, foreseeable privacy and notification risks, withdrawal procedure and contact point for questions. Obtain any required institutional/ethics approvals before starting human-participant data collection; maintain a documented access and incident-response plan. Evaluate notification burden and sleep-related anxiety as possible harms, and stop or revise the intervention if user welfare is adversely affected.

6. References

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